

nodes simultaneously and change the blockchain such that their version becomes the majority copy and the agreed-upon chain. Because of the new timestamps and hash codes, all the blocks would need to be redone, which would demand a huge investment of time and money.

Considering the massive scale and explosive growth of many cryptocurrency networks, the resources required to accomplish such a task are likely to be beyond the reach of any human being. The cost involved would be prohibitive, and the results uncertain. It would be impossible to conceal such a massive change to the blockchain from other network users. Afterwards, the nodes in the network would fork off to a new, unaffected chain version. This would lead to the value of the token being targeted plummeting, rendering the attack useless since the bad actor would have a worthless asset. The same thing would happen if the malicious actor attacked the fresh Bitcoin split. This design makes contributing to the network more financially beneficial than attempting to disrupt it.

Forms of blockchain technology

A total of four distinct blockchains exist today.

Private blockchain networks:

Private blockchains are only accessible inside a restricted network, making them ideal for use within exclusive companies or groups. These blockchains allow businesses to regulate user access, set network settings, and implement other critical security measures. In a single-administrator blockchain network, just one person is in charge.

Public blockchain networks: Public blockchains, the technological basis for Bitcoin and other cryptocurrencies, have also contributed to the widespread use of distributed ledger technology (DLT). For example, the problems of centralisation and security breaches are less of a concern thanks to public blockchains. Instead of keeping

information in one central repository, DLT spreads it out over a network of computers. Proof of stake (PoS) and proof of work (PoW) are two popular consensus algorithms used for authenticating data.

Permissioned blockchain networks:

Permissioned blockchain networks, also known as private or hybrid blockchains, only allow access to verified users. This hybrid blockchain setup provides organisations with the best of both worlds by allowing for more precise control over who has access to the network and which transactions they can take part in.

Consortium blockchains:

Consortium blockchains, like permissioned blockchains, include public and private features. However, unlike permissioned blockchains, a consortium blockchain will be managed by numerous entities. These blockchains may be more difficult to set up initially, but provide enhanced security once operational. To add, consortium blockchains are best for working with several different companies.

Exchange methods

The confirmation and authorisation of transactions is a crucial component of blockchain technology. If two people, each with their own private key, want to conduct a transaction, the first person affixes the details of the transaction to the second person's public key. All of this data is compiled into one big chunk.

The block has crucial data, including a digital signature, a timestamp, and more. It does not record the names of anyone who participated in the transaction. This block is then broadcast to all other nodes in the network, and the transaction is completed after the appropriate user validates it using his/her private key.

The blockchain is not limited to only handling monetary transactions; it may also record sales of real estate, automobiles, and other assets.

The pros and cons of blockchain

Despite its seeming complexity, blockchain has almost limitless use as a decentralised ledger system. Blockchain technology may find utility in a wide variety of contexts, including (but not limited to) the following:

- Improved user privacy and security
- Reduced processing costs and mistakes
- Streamlined operations

On the other hand, there are a few drawbacks to consider.

Benefits

- Better accuracy since no humans are involved in the checking process
- Reduces expenses by doing away with independent verification
- It is more difficult to tamper with anything that is distributed
- The transactions are safe, confidential and quick
- Uses open source software
- Gives people in nations with shaky or undeveloped governments access to financial services while protecting their privacy

Drawbacks

- Bitcoin mining requires a lot of expensive hardware
- Slow rate of sales in a certain period
- A long history of illegal usage, particularly on the 'dark web'
- There is no universally accepted set of regulations, and regulatory differences exist among jurisdictions
- Data storage capacity constraints

What the future holds

Since blockchain is a relatively new technology, opinions on its future are still divided. According to a survey conducted by *TechRepublic Research*, 70% of working professionals have yet to use blockchain. However, 64% state they believe blockchain will impact their sector, and most anticipate a favourable outcome.